

Project Initialization and Planning Phase

Date	24 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Choose the most suitable crop for my land to achieve a better yield.	I am not sure which crop is suitable for my soil and weather conditions.	I don't have enough knowledge about soil nutrients, rainfall, temperature, humidity, and pH levels.	Worried about crop failure and financial loss.
PS-2	An agricultural researcher	Analyze how soil and climate conditions affect crop growth.	It takes a lot of time to analyze large amounts of agricultural data.	Manual analysis is slow and it is difficult to identify useful patterns.	Frustrated due to delays in research and decision-making.
PS-3	An agribusiness manager	Recommend suitable crops to improve agricultural productivity.	Crop recommendations are different for every location and season.	Soil and weather conditions keep changing, making manual recommendations difficult.	Concerned about providing accurate crop recommendations.
PS-4	A government policymaker	Promote sustainable farming practices.	It is difficult to provide personalized recommendations to farmers.	Existing systems do not use AI to analyze soil and environmental data effectively.	Motivated to adopt smart technology for improving agriculture.